

1. Problem Statement

Door assembly is a critical bottleneck in automotive manufacturing. Current operations rely on manual part logistics, unpredictable supply arrivals, and labor-intensive station setups. High defect rates due to misaligned parts, missing fasteners, or incorrect door variants (LH/RH, trim levels) cause rework, line stoppages, and warranty claims. Peak production targets are rarely met due to assembly delays and supply chain inefficiencies. Traceability gaps lead to quality escapes and difficulty root-causing assembly failures.

2. Root Cause

- 1) Disconnected Parts Logistics:- Door panels, hinges, window regulators, and trim components arrive through separate suppliers with varying lead times. No automated tracking; parts pile up in staging areas or arrive late, causing assembly delays. Line workers manually hunt for parts.
- 2) Manual Door-to-Variant Matching:- Line supervisors manually verify that door panels match vehicle specifications (LH/RH, color, trim level). Mismatches discovered late (post-assembly or post-paint) trigger expensive rework and line re-runs.
- 3) Labor-Intensive Station Setups:- Technicians spend 15–20 minutes per shift repositioning jigs, swapping tooling, and verifying part kits for different door variants. No visual guidance or pre-assembly part staging.
- 4) Poor Defect Visibility & Traceability:- Quality escapes (missing fasteners, misaligned hinges, water seal failures) traced back to line-of-origin only after customer complaints. No in-station defect capture; rework happens off-line.

3. Solutions

- 1) Automated Parts Supply Chain Synchronization:- Integrate supplier order feeds with production schedule. Use IoT tracking on inbound parts containers (RFID/barcode). Automated parts sorter routes door panels, hinges, and fastener kits to station-side buffers based on real-time assembly schedule. Eliminates manual part hunting; ensures correct variant is staged before line arrival.
- 2) Vision-Based Door-Variant Verification & Kitting:- Computer vision system scans door panel markings and color at incoming inspection. Cross-references against vehicle build spec from MES. Automatically routes to correct assembly station and flags mismatches immediately. Pre-assembles and presents correct fastener kit per variant.
- 4) In-Station Quality Checkpoints & Real-Time Traceability:- Add vision/force-sensing stations post-assembly to verify hinge alignment, fastener seating, and water seal placement. Capture defects in-situ; log to MES with operator, timestamp, and part serial numbers. Feed defect data to quality analytics for root cause and supplier accountability.

4. Impact/Result:-

- 1) 25–35% Reduction in Door Assembly Cycle Time:- Automated parts supply + jig automation eliminates manual hunting and setup delays. Average assembly time per door reduced from 8–10 min to 5.5–6.5 min.
- 2) 40–50% Reduction in Assembly-Related Defects:- Vision verification prevents wrong-door mismatches; in-station checks catch alignment and fastener issues before handoff. Warranty claims from door assembly drop significantly.
- 3) 60–80% Reduction in Rework Hours:- Defects caught in-situ instead of at final inspection or in field. No post-paint door re-work; no customer returns for door-related issues.
- 4) 20–30% Increase in Line Uptime:- Elimination of bottleneck pauses due to missing parts or setup delays. Fewer line stoppages for defect resolution; production schedule more predictable.
- 5) 100% Parts Traceability:- Every door panel, fastener kit, and assembled door linked to vehicle VIN, operator, station, and timestamp. Full audit trail for quality, recalls, and supplier performance.
- 6) Data-Driven Supplier & Process Improvements:- Defect and parts-arrival data surfaces supplier quality trends and lead-time patterns. Enables targeted negotiations and process optimization at supplier level.